Livestream 058.2: "From pixels to planning: scale free active inference"

1:59:21

hello and welcome everyone it is
November 13th
2024 we're live and kicking off the
fourth applied active inference
Symposium it will be a great program and
we will begin with Carl friston and
Lance
dcosta discussing scale free active
inference we discussed in live stream
58.1 one with Lance last week and in
58.0 with Aron two weeks ago a bit about
this work and today Carl thank you so
much for joining and presenting on it so
looking forward to it and throughout in
the live chat please feel free to write
any comments thanks again everyone for
joining and looking forward to
this well thank you Daniel it's a great
pleasure to H kick off the Symposium
this year um I'm going
to rehearse um a
presentation I made at the um active
inference Workshop um a few months
ago but this time I hope in a slightly
more relaxed uh and unpack way so I'm uh
fondly hoping for lots of questions and
uh embellishments from the uh from the
audience and uh and from Lance um this
is a bit of a colloquial presentation
it's just going to focus on something
specifically that we've been working on
for the past few months which is
effectively addressing the question how
can one scale active inference for high
dimensional real world problems and the

solution that we've been pursuing is to adopt a scale-free approach um this work has been written up um and put on archives
um however it's also been submitted for uh publication peer reviewed publication to a special collection um in uh memory of Herman hen um and we've equipped the peer reviewed version with a little forwards I just wanted to frame or preempt the scientific part of this presentation just by motivating why why the work of Herman hen is foundational and under writes many of the
um ideas and procedures that underly this scale-free approach to active inference so Herman hen was famous for synergetics um a physics Le approach to self-organization um and in particular the importance of understanding the coupling between different scales of Dynamics in self-organization
and if you go to the Wikipedia entry you will read about macroscopic and microscopic variables and how they are coupled to each other in a in the fashion of circular causality um so there's both bottom up and top down causation where the top down causation is sometimes read in terms of um the slaving
principle I'm going to present this approach um to harnessing this um notion of a scale-free universe uh in terms of installing that scale free aspect in the generative models that we bring to the table to solve decision- making to uh simulate or um emulate active inference

under discrete State space models
so let me start at the beginning um the
nature of things and the free energy
principle um and this is just to
rehearse the fundaments of the free
energy principle it all rests upon the
notion of a Markoff blanket where we can
separate internal and external states by
boundary States or states that
constitute the mark of blanket namely
the sensory States and the active States
and for those people not familiar with
um this particular partition of states
of any random dynamical system it's just
specifying systemically um a system say
a brain in terms of its inputs and
outputs and um just
formalizing the conditional dependencies
that Define what is an input to a brain
or an agent and what is its output in
terms of the sensory and active states
that thereby render the system of
interest in this instance a brain open
to the environment the econ Nish um the
heat bath if you're a physicist or
chemist um in virtue of the fact that
the inside can influence the outside via
the um active States the outputs and the
outside can influence inside via the
sensory States or the um or the inputs
and with this
setup we can elaborate um a kind of
physics namely uh um basing mechanics
that is almost well formally um related
um and inherits exactly the same
mathematical and functional form as all
the other mechanics that we find in
physics including quantum mechanics sta

statistical mechanics and classical
mechanics the the only thing that
separates the um the basy mechanics is
the fact that we have in mind explicitly
this partition between internal States
and external States and the uh blanket
states that intervene and couple the
internal to the external and that
basically mechanics is is effectively
summarized here in terms of um
variational and expected free energy um
where we can read the expected free
energy as effectively PRI beliefs about
the kind of outputs or actions that um
in this instance a brain could exert on
the external States um and the
variational free energy keeping
implicitly the um the internal States a
a good model so it's tracking in a
probalistic sense the um the Dynamics on
the outside and we can combine this to
uh create a a generalized free energy um
that now te logically or or perhaps even
an anthropomorphically we can now read
the gradient flows that subtend this
phasing mechanics in terms of action and
perception just by
identifying the internal and active
states of any agent as autonomous States
and noting that they effectively um um
flow on the same free energy functionals
so that's the basic setup um but what I
want to do before um applying the free
principle in the service of things like
active inference is just back up and
reflect upon the fundamental role of the
Markoff blanket and how that leads to a
scal

invariant picture or view of Dynamics
and self
organization so what I'm going to assume
is that ultimately we will assume that
the basic mechanics um means that the
Dynamics of anything that is defined in
terms of its boundary or Mark of blank
it can be described in terms of active
inference but before that we have to
think about the nature of things um and
in particular
um if one goes back to the particular
physics paper um the nature of the
states that constitute things or
particles um and there's a bit of a
subtlety here because before you can
start talking about States you really
have to Define what is a state is it the
state of something is it the state of a
particle um and that presents an issue
because you're defining a part iCal in
terms of this particular partition into
internal external uh and Boundary or
blanket States so there's a um um
there's a slight toor not topology but
there's a slight um problem in resolving
how you define states of things or
particles and particles that possess um
States and the resolution of that is to
adopt um a recursive definition so that
the particles constitute a set of
microscopic blanket and internal
States while the states are a set of
macroscopic igen functions igon modes or
just simply mixtures of the blanket
States so I'm here deliberately um using
the notion of microscopic and
microscopic States just in uh as a um a

nod to synergetics and particular kind
of synergetics that um Herman hen
developed and subsequently uh championed
so this uh is cartooned or Illustrated
um on the right hand side here so we
sort of take some states um and um uh
and then um apply a particular partition
on the basis of which states are coupled
to which other states and the
conditional independences that
ensue um and then having done that we
can then take mixtures zon mixtures of
the boundary states that then generate
new States um and then we can um
effectively by taking those mixtures we
effectively reduce the dimensionality so
that the higher scale um coarse grains
the lower scale and then we reduce it
and then we just rinse wash and repeat
and we do that and every time we sort of
pass through this cycle of RG operators
um reduction and course graining
operators we move to one scale above
successively and this is the scale
invariant aspect of um that I want to
leverage in in the remainder um of this
presentation um I use the word RG uh
deliberately because um one way of
looking at this or articulating this is
to use the apparatus of the
renormalization group so this is where
renormalization gets into the game that
you're recursively um applying these
operators Co graining things um in a way
that conserves their Dynamics and that's
an easy um um uh that's an easy criteria
to meet for us because we know that
anything that has a particular petition

can be described in terms of a gradient flow on variation or generalized free energy so we have everything that we require in order to apply um the apparatus of the renormalization group so that what that basically does for us once we recognize that
particles are composed of states where the states are ion mixtures of particles in this scale invariant fashion gives us a view of any universe that comprises things of things of things
um as um inherently uh resolving a chicken and egg problem through scal invariance by application of the renormalization group so here's just another illustration of how that actually works in practice when you write down the equations I'm not going to bother going through the equations this is just used iconically um just to make the
observation that as you move up scales um so for example in an evolutionary context this would be the sort of the Dynamics of phenotypes and then we move up to the an evolutionary scale by um effectively Co graining in a way that Herman hen um would exactly prescribe in terms of taking those slow macroscopic variables that decay very very slowly um which are technically the igen mixtures equipped with very small um igen values whose real part approaches zero From Below effectively saying that these patterns Decay very very very slowly and they have a kind of persistence that lends higher and higher

scales a temporal dilation so things
change more and more slowly as you
increasingly uh cor grain things and
that this falls out of the maths um when
you simply apply a particular partition
and put that partition I in um as the um
the course grading operator in this uh
in these in this scale free
formalism so in short basy mechanics
entail a generative model of external
Dynamics but we've just seen that just
by defining things in terms of particles
that have States and defining States in
terms of the ion functions of particles
that external Dynamics are inevitably
scal inv variant it can be no other way
and therefore the generative model must
be scale invariant or when expressed as
a graphical model scale free and I just
slip that in as a
um a nod to the fact that people often
talk about scale freeness um generally
speaking when people apply the notion of
scale in variance to anything if that
thing is a graph then you can call it
scale free so a scale free thing is a um
is a graph that has has this scal inv
variance crucially in both time and in
space because we are dealing with
dynamical systems and we're doing this
um Dimension reduction or course
graining um using these um migen
functions that necessarily pick out
these slow macroscopic sometimes called
stable modes they're stable because they
they don't Decay or dis ipate um almost
infinitely quickly so they're stable in
the sense that they Decay very very

slowly so how can we harness that
observation that truism at least truism
under the world as described by hen um
in um the construction of plausible
generative models and in particular um
generalized Markoff decision or
partially observed Markoff decision
processes um well what we can do is we
can just look at the different kinds of
depth um and ask at what point would we
um be able to leverage um this
renormalizing aspect of the cause effect
architecture of Any Given world that
we're trying to navigate and predict so
here's uh the normal um depiction of a
um
a a generalized um Mark of decision
process and by generalized what I mean
is that there's um an explicit
representation of not just the states of
things that are generating outputs or
things that can be measured or observed
on the um on the sensory sector of a
marov blanket but also their Dynamics in
terms of the paths that um specify the
transitions between state
as we know normally encoded in terms of
a transition tensor be here um so just
by putting in the paths as explicit
random variables that accompany for each
given Factor um the hidden States we
have a generalized kind of uh Markoff
decision process uh encoded in the usual
way in this instance um in terms of
tensors um that themselves can be Ariz
to distributions so um that
parameterization equips these kinds of
models with a parametric depth in the

sense that if something is deep
parametrically that means that the
parameters of various probability
distributions are themselves random
variables that have their own
probability distributions so for example
the likelihood mapping that maps from
sensory states to outcomes here uh
usually encoded by um an a tensor here
mapping from latent states to
observations is itself encoded in terms
of um diset parameters um blending this
uh kind of model A parametric depth and
um one can ask what the implications of
that are well the implications are
another kind of depth in terms of um you
now have to optimize or the B apply
basic mechanics to the latent States the
hidden state and that would be inference
and then you also have to um consider at
a slightly slow time scale the
optimization of the derish parameters
and that would correspond to learning
and so on um if I wanted to go to the
structural learning of the model in of
it in and of
itself we also have um implicitly a
temporal depth in the sense that we're
going to be raring out into the future
and that there are dynamics here um and
this Temple depth is generalized in in
virtue of having um these explicit uh
path variables that we're going to uh
exploit later on we can also have
hierarchical depth in the sense that we
can put
um compose many markof decision
processes um um in the in the sense that

the outputs of one process now
constitute the inputs or the empirical
prior the top- down constraints the
induct biases on the Dynamics and states
of the level below and the Dynamics are
um um the kind of inputs or sorry
outputs of the uh upper level that
provide empirical PL on the paths and
the initial conditions are supplied or
mediated by a mapping between higher
levels slower States um and the initial
conditions or initial States of the
lower uh of the lower level by this D
tensor here that you can lump together
um in terms of sort of high you can lump
together the D and the E tensors they
play the role effectively of the a
tensors in sort of um mapping from one
level to the uh the next level there's
also factorial depth which many people
will be familiar with in terms of um for
any one of these um
lumped States here there may be many
many different factors that have an a
certain Independence structure such that
you have interactions between factors in
determining outcomes um you can look at
this in terms of uh agative model when
equipped with factorial depth
effectively entangles latent states to
generate um outputs or outcomes which
themselves can be prior constraints on
the states and dynamics of the of the
level below such that the inversion of
these models effectively disentangles
the data the content that is being
observed to explain it um in terms of a
disentangled representation which is the

um the latent State and dynamics of the generative model so what we have here then is um a generalized discrete State space model with Paths of as random variables um just to um return to this um fundamental aspect that as one ascends scales in real life in um you in terms of uh the external States things slow down in virtue of the um the the the the synergetic um mapping from Fast microscopic States uh to um slow macroscopic States um the same thing can now be implemented in these um markof decision processes that have a hierarchical depth um and the way that is done is effectively to discretize or quantize time such that there there are more updates at any given level than any um than the level Above So for every um update at this level there are uh say three updates or two updates at the level below and that um affords a temporal scaling um in accord with the scaling variance um implied by the synergetic view of self-organization I've tried to cartoon that here in terms of if you imagine lots of these um little markof decision processes populating this uh circumference here and that as we go deeper and deeper and deeper into a model with hierarchical depth as we consider updates in Universal or clock time there are more updates at the peripheral level exposed say to the the um external states of entire brain um relative to the relative to the updates as you get deeper and deeper and deeper

in into the
model so
where the temporal depth um becomes if
you like intimately tied to the
hierarchical depth the higher the
hierarchy the slower the updates which
is why um we generally parameterize the
initial conditions um or generate the
initial conditions and the initial path
from the states above and that path can
pursue um um a trajectory over multiple
time steps until a certain um Horizon uh
and which case it then sends messages
back to the level above likelihood
messages um that then cause belief
updating um in the subsequent Step at
the level above and then generating
priors about the subsequent path the
next PATH at the level below and then so
on recursively uh to the depth of the
model um so that's the time bit what
about the spatial bit what about the
spatial core graining the lumping
together of various States um and of
course this depends upon the particular
partition so in the General Physics
formulation you'd be looking at uh
Markoff blankets and markof blankets in
Practical applications one can take a
shortcut um and that shortcut
effectively rests upon um the
simplifying assumption that for most
systems that are um composed of um
coupled um coupled uh dynamical systems
um the interactions are usually local so
if we um just think about um some um um
system that um can be described in terms
of local interactions so for example um

in a um a ukian world um that comprises some ma massive objects that bounce around and if we ignore gravity for the moment then they're only going to influence each other or depend upon each other when they're touching when they're um when they're proximate um effectively what that means is the the the mark the particular partition that defines all the Markoff blankets in um in this particular State space um um is composed of lots of local little tiles um so what we can do is we can use a um a device um very prevalent or indeed ubiquitous and applications of the renormalization group which is called a uh a block or a spin operator that just groups together local tiles or quadrants of a say two-dimensional arrangement of time of of States so I've Illustrated that here just by um in terms of um How We Do the spatial cor graining in these um what will refer to as a renormalizing generative uh model with spin block uh Transformations um in terms of um this little um um this hidden state or hidden factor with its states paths here um is responsible for generating predictions about the Dynamics of these two um factors um at the um at the lower level and in so doing what we are effectively doing is successively grouping together sets of sets of sets of sets so that any set or group at one level is accountable for or trying to generate the Dynamics of a small local group at the at the lower

level um this is quite fundamental this
kind of Architecture is quite
fundamental because what that means
is anything at the lower level only has
one
parent and if it only has one parent and
if you remember the markof blanket is
constituted by the the the um the parent
and the children the parents of the
children because there are no co-parents
there are no parents of the children and
what this means practically is that we
can now ignore dependencies between or
effectively resolve or dissolve any
dependences between the factors at any
given level because each factor is only
responsible for its children and it
doesn't share any children in virtue of
this um particular spin block
transformation so this leads to um a
very efficient kind of generative model
um well certainly an efficient kind of
inversion of the gened model um where
we've got all these converging streams
but the streams don't have to um uh talk
to each other until at the point that
they come together through these
blocking or spin block
um Transformations so at some point all
the tiles or the groups at the lowest
level will actually come together in um
uh at the top or very deep in the model
at which point we're now encoding quite
long trajectories technically 2 to the^
of the depth minus one um in terms of
updates um but at the lower level um
there are no interactions between the
different blocks as we Ascend them and

this is just to um remind you we can
treat these D and E the um the initial
States and the initial paths effectively
as likelihoods that couple um one scale
or level to the scale
below so how uh with if that's the kind
of model that we aspire to because we
think that's appropriate for any world
that shows this scale uh invariant
aspect in space and in time or in state
space and in time um how are we going to
build one of these things
um and this is the sort of second um if
you like
um thing that we have been working on um
um because it's going to be very
difficult to specify by hand these um
scale-free or renormalizing generative
models we really want the model to learn
itself so basically how does one build a
renormalizing generative model and what
we're going to do we're going to use
very fast structure learning and this
particular kind of structure learning is
necessarily recursive and uh in the
sense that we're going to build these
models through the recursive application
of these um RG operators or
renormalizing operators so what's fast
structure learning so fast structure
learning basically equips the model with
unique states and transitions as they
are encountered so to put that simply
what we're going to do is grow a model
from scratch from nothing and how are
we're going to do that well we're
effectively going to take one
observation and say okay that was

generated by some latent State and then we're going to take the next observation um and say okay I haven't seen that particular observation before um so I'm going to induce a second latent State and furthermore I've seen these two of observations in sequence therefore I'm going to um encode that with a transition from the first to the second state and then I just repeat I take the the third one and if I haven't seen that before I induce a third latent State and a third a second transition um accumulating all the Unseen or unique instances or um um of observations in a likelihood M mapping that just grows and grows and grows until we have seen something before so we don't induce a new state or implicitly column of the likelihood Matrix if we've seen uh the previous state that's just a um um an instance of something that has been seen before um it may well now come along with a different uh path so the next state may not be the same as the the subsequent State following the previously scene instance of this state and therefore I'm going to induce A New Path and slowly grow my B tensors and my probability transition matrices until I can um basically encode all the dynamical structure of the effectively the training sequence of observations summarized in a a maximally efficient way and I mean maximally efficient in a technical sense um in the sense that the implicit map that we are growing from

from scratch um has the highest Mutual
information between the latent States
and the observations that are at hand um
the way that you can think of that in
terms of um free energies that the
expected free energy um is just the
mutual information in the absence of any
expected cost so the expected free
energy is just the mutual information of
any mapping um um
um plus um or minus an expected cost
where cost is defined the usual way in
terms of constraints or prior
preferences
so one can actually look at this uh fast
structure learning um as a particular
kind of active inference but not about
um it's not active inference Over States
or Active Learning over parameters it's
active selection of the right model in
terms of the size of the likelihood
tensors um and the transition uh tensors
um and the criteria for whether we
accept a new latent state or a new
transition is simply does it optimize or
improve the expected free energy namely
does it preserve or conserve or maximize
the the the mutual information so we can
actually look at this structure learning
as just another instance of act
inference but here deployed in terms of
um um in terms of basic model selection
or some some people call this uh
structure learning that has exactly the
same rules as um planning um and um
uh and um accepting um parameter updates
provided they they maximize expected
free energy as well um the applic ation

of this kind of uh fast structure
learning to um these renormalizing
models U looks complicated here's the
algorithmic description of it um again
please don't worry about the details
here it it's relatively simple you know
all we're doing basically is ingesting
um observations but we're just doing it
for local groups um and then we are
combining the initial conditions and the
paths into new outcomes for the next
level um and so because we are doing
this for local groups the number of
groups just get smaller and smaller and
smaller as you as you get higher and
higher and higher but obviously the
number of combinations of paths and
state initial States increases as you
get deeper and deeper into the model and
you may be asking well you know how do
you ban that you know how can you um
possibly assimilate all the data without
having an extremely large number of
latent States very deep in the model
that are now encoding effectively
evolving patterns or paths over quite a
large number of time points into the
future well it's trivial to do that
because the size of the highest level
I'll call them generalized states that
basically generate paths and initial
conditions for the lower level
um is upper bounded by the number of
elements in your training set um so that
you can control and upperbound the size
of these
renormalizing operators or likelihood
mappings and transition um um tensors

simply by giving it canonical training
data the kind of data that you want this
um agent to generate or this um um agent
to recognize
and thereby
predict um so here's a a particular
example of that um what we've done
here is just ask can we very very
quickly learn a generative model that
will generate a um a simple movie and
this movie is simple because um it
begins and ends in the same state so it
just Loops over and over and over again
um so we only have to supply one Loop um
to this fast structure learning so that
the model builds itself and can then be
used in generative mode to generate the
movie time and time and time again um so
here's one depiction of um the results
of the learning the the F structure
learning um and subsequent inference
when exposed to um this kind of image so
the coarse graining here at the very
bottom in in fact um uses um um a
slightly finessed version of the spin
block Operator by um harvesting um local
igen modes or igen ion images um of
little patches of of the video to get it
into a reduced or coarse grained
representation that then is just passed
up using these spin block operators and
in this in we just need uh two levels to
the model to uh have the highest level
see the entire image and then if we
present the image um um after
discretization using the singular
decomposition or the sequences of images
to a model it learns the model and then

we can generate um we can generate um
little movies simply by moving through
these paths or episodes I use the word
episode um
because remember these generalized
States the highest level encode paths or
trajectories into over um 2 to the N
minus one time steps in the future when
n is the um is is the uh the depth of of
the model and these episodes themselves
have Dynamics so we just go through a
little Loop here go back to the
beginning and thereby generate um this
image at the
bottom what else can we do with these
kinds of models
well one thing we can
do is
um present not the entire data but just
a little prompt um
so in this example what we've done is
actually show it just the upper right
quadrant of the
movie and the and see what this um and
this generative model um believes is
going on in terms of the um the
predictive post area in output space in
image space in this
instance now because during training
it's only learn to recognize a dove
flapping her
wings um that's all it can believe can
is the cause of its Sensations even if
those Sensations are very partial so
basically what we have here is a kind of
completion that inherits from the fact
that this generative model only knows
about latent causes that constitute an

entire Dove flapping her wings so um
even though this is what the dove is
actually um exposed to it's sensing this
is the predicted after a couple of
frames it very quickly um is able to
predict um what it would what it should
be seeing or what it would have seen had
it been given the entire uh data and
another way of thinking about this um
remarkable ability to sort fill in the
gaps um is from the point of view of
compression so just a little digression
here
um a lot of the basy mechanics and um
variation inference that uh ensues from
variational mechanics can also be
likened to um
the to
um the same fundaments underwrite um um
efficient information transfer so we've
talked about maximizing Mutual
information in the context of uh um
algorithmic complexity and um the
associated Universal computation um then
what the objective function looks like
is basically how much can I compress
this content um to represent it in the
most efficient way so you can look at
this past structure learning under these
renormalizing or scale-free generative
models as the most efficient way of
compressing whatever your data you've
been exposed to um and because we've
compressed it in such a way that it can
only now recognize a dove flapping her
wings um then it can if you like fill in
the gaps in a very efficient way when
it's given partial or noisy data and I

use the word noisy here because you can
regard the missing three quadrants here
as extremely imprecise data data that
has extremely low signal to noise for
example so um here's another example in
the previous example we we um took a um
effectively uh
a movie or a dynamic or um a system that
had a periodic orbit um um which was
each flap of the Wings will this work
for um a periodic orbits and and
generally for um stochastic um uh chaos
um yes it works it works relatively
efficiently so this example um we um
used training data that was generated um
um through creating images of a white
ball that moved with a chaotic
trajectory governed by a uh a loren
system so this is the kind of system
that governs or is used to describe
chaotic events um in meteorological
systems or in fluid convection
um so by applying exactly the same
procedures as in Pre in the previous
example um we now uh generate this
instance a three-level um um hierarchal
or renormalizing model um that now
accommodates the um stochastic or
Randomness induced by not only random
fluctuations on the Dynamics of this
stochastic uh random dynamical system uh
but also if there were any deterministic
aspects the deterministic uh um um chaos
that ensues from exponential d
convergence of trajectories that now has
been summarized stochastically in terms
of probalistic um uh transition matrices
here which means that um we can having

compressed the training data we can now
withhold not a quadrant but just simply
withhold the data to stop the data uh
but the model will keep on generating um
what it predicts would have happened
given the statistics of the Dynamics
that have been in store
in this renormalizing generative model
and what we've done here is um um for
the first um few hundred time bins at
the lowest level we basically rendered
the images or the input very very
imprecise so that everything is governed
now or everything is generated from this
top- down Prize or posterior um um
posterior um predicted posteriors that
inherit from from precise PRI is because
there's no precise likelihood um and
then we've actually uh removed the U um
removed the um imprecise input
completely and just allowed the um the
agent to generate its own beliefs about
what's about what's going on um so here
this is fairly precise in the first
instance and then we remove completely
the um stimulus but it still uh imagines
or generates its or predicts
um what's going on in terms of this
chaotic um Dynamics um and then um with
a degree of uncertainty then um
continues to then generate um the um the
Dynamics associated with this kind of
stochastic
chaos here's a pretty example um using
exactly the same uh technology and
approach um but here explicitly um
talking about um video compression um so
we literally just presented a video of

naturalistic Dynamics in this instance a little bird a robin um feeding um itself um and um compress that using these uh this renormalizing um generative model um such that we can again simply remove stimuli at certain points um in the video um and yet the the in in the agent's mind nothing has really happened because all it knows is that if it was like this um previously then this must have happened um and so on through this in this instance um orbit that doesn't have a a closure it's just a trajectory through from the beginning to the end of the sequence and here's the postor prediction after just being exposed to a couple of frames right at the beginning um in the first um six uh time frames of um of about 12 sorry of um 18 um this example moves applies exactly the same um approach um as previously but now we're not talking about um ingesting or learning or compressing images we're talking now about compressing or ingesting sound files uh um and that is if you like a simple or limiting case of this two-dimensional uh compression that we talked about before so in this instance now what we're doing is compressing one-dimensional frequency summaries of a particular um wave file or a um file um that um describes fluctuations and the intensity of uh of a sound that then is um summarized into terms of a continuous wavelet transform um and then the succession of different trajectories in this um time frequency

space is then ingested or assimilated or
learned by the generative model um now
unfortunately you can't hear this um so
I'm not going to bother playing it but
um it um this is what the agent was
trained on it's basically J
piano um and these are the um ensuing
predictions in the absence of any inputs
it's been used to generate music um
using um the stochastic
transitions um through various bars of
Music um that it has learned um
returning to one or other bars of the
munic of the music when it reaches uh
the end of the orbit to generate
something that um sounds at least Jazzy
if
not oh no
started it but you can't hear that so
I'm not going to pursue that so to
summarize what we've done so far um
we've gone beyond generative AI in the
sense of um
um having a a true generated model um uh
that entails active inference in agency
um and in the setting of these
um well sorry what we are going to do
now is go beyond just simply generating
uh content um having been uh trained on
the basis of some training data um and
we're going to now consider um agency by
putting action into the mix um and in
the setting of um the simplified setting
of optimal State action policies where
we can effectively um ignore the
imperative to maximize expected
Information Gain or um um
explore in the sense of uh responding to

epistemic affordances um we can
um do one of two things we can either
use some Universal function approximator
say deep learning to optimize the
mapping from sensory to active states to
maximize the um expected cost or the
expected utility part the expected free
energy um or we can use active inference
to realize predictions under enormous
izing generative model of rewarded
events so what I'm doing here is
deliberately setting up a dichotomy that
tries to distinguish the active
inference approach to purposeful
Behavior where the purpose is provided
purely by the prior preferences or the
um um or the the uh the constraints
usually encoded in the c um uh tensor um
against a sort of reinforcement learning
approach and this is rather busy slide
but um I think it's probably worth just
um uh briefly uh rehearsing the
distinctions between active inflence and
reinforcement learning so the you know
the story with respect to action as a
gradient flow on free energy which
basically reduces to um picking those
actions that maximize the predictive
accuracy um in the sense that if you
look at the expression for the variation
free energy the only thing that you can
change um is the accuracy in the sense
that free energy can be written as
accuracy minus complexity minus accuracy
and the complexity um is does not depend
upon action whereas the accuracy does so
that picture basically says um I can
describe behavior in terms of a

generative model that um is optimized
through um perception uh in the usual
way um cast in terms of say variational
inference um that basically generates
predictions about what should happen
next and then I can pick my actions to
fulfill those predictions in continuous
state for formulations this would be
without minimizing the prediction error
usually the proceptive domain but you
can generalize this to any any
consequence of action I'm just going to
pick that action that brings about or is
most likely to bring about what I
predict will happen next and what I
predict will happen next um is optimized
through perception through variational
inference based upon my learned and
selected um generative model so this is
very similar to sort of um model
predictive control um and neurobiology
could be cast in terms of um um ual
control theory
um which puts action in the game
basically of
realizing
evidencing um predicted States so I've
tried to summarize that here just in
terms of um you know our Markoff blanket
where we're optimizing our beliefs about
the world um noting that planning in
this instance so um planning as
inference uh is all about the path it
does not have to be about action it's
just about how does this world
unfold um and from the point of view of
these scale-free generative models we're
really thinking about how do episodes

follow each other which episode is going to follow this going to um um this episode and we can say well you know you are in control of that um you know imagine a um a benevolent World in which everything happens according to um your beliefs about how to be an expert in this particular world and the agent can plan a series of episodes generating predictions all the way down to the bottom which then slavishly action can fulfill just by fulfilling those predictions um so one could in some sense uh think of this in terms of controller's inference at the bottom level in the spirit of mod predictive control and planning as inference on the top level so belt and braces in terms of realizing active inference in this particular example where I repeat um we are really just worried about um State action policies um why are we worried about State action policies well these are the kind of policies that reward learning schemes um are aimed at um and have a similar architecture um but um to my mind a slightly more um um engineered architecture first of all you actually need explicit inputs which are rewards um there are some proxy for um the constraints uh you know um um under which um the basic mechanics would normally operate um but more importantly you're you're you're basically sending um your inference scheme is basically about selecting the right policy I've used Q learning here um um to maximize the disc counted reward um so

similarities and differences but we're
going to focus on uh active inference um
uh in this in the particular setting of
these renormalizing models where repat
planning and thinking or imagining a
future can proceed at the very highest
level that has no notion of action
they're just patterns that unfold they
may or not be caused by me I don't know
I don't care um but the action at the
lowest level the refle Es as it were um
do um are in a position to make my um
post predictive posters come true simply
by picking those actions that realize
the predicted outcomes
so how are we going to leverage that
kind of merely reflexive active
inference um in the context of building
or learning from scratch um
models that would be apt for doing
something implementing a state action
policy um and I'm going to um appeal
here again to this notion of um active
selection as one way of um looking at um
action which can always be at least phys
mathematically expressed in terms of
um of
um doing something if it minimizes
expected free energy um and in this
instance um what we can do is actually
apply it not to the selection of the
model but the selection of the training
data so if we just um apply
the the dictat that you only select
something if it um um minimizes expected
free energy the
includes um the expected cost what that
basically means is I'm only going to

select those data features are those
training sequences that don't come along
with a high cost namely those that are
rewarded um and that's what we're going
to do here we're basically going to um
learn to play um a very simplified um
game of pong where this ball bounces
around inside a circle um
and we can move the bat um in a way to
uh return the ball so that we can avoid
penalties when the ball hits the upper
boundary um and secure a reward um
whenever um those contact of the ball
and the bat and how can we learn this
kind of dynamic these episodes in
exactly the same way we learned the
video of the robin or the the um the
Dove flapping her wings well what we can
do is we can just um select um instances
of random play that conform with um our
um dictat that we cannot um include
those costly events and so what we've
done here is just generate training data
training
sequences using random
play but om
anything that does not that contains a
costly outcome and furthermore just to
make things more efficient we've
restarted the random play after a reward
is secured and then just use the
subsequent play if it actually attained
a subsequent reward so basically
chaining together bootstrapping together
or splicing together sequences of
rewarded uh play um and then use that to
um uh to um submitting that for fast
structure learning and now we have a

generative model of expert play
effectively or a generative model that
has that entails a state action policy
that can be realized through um um
reflexive active inflence so active data
selection um and I'm um sort of
cartooning that here in terms of
applying a certain kind of Maxwell's
demon that only lets through non-costly
episodes um basically yes admit the
sequence for fast structure learning if
there is a reward if not ignore this
ignore this sequence um and this
effectively learns um or furnishes a
generative model events that lead to a
reward and nothing else so now this
agent can only recognize expert play and
therefore all of its predictions have to
be uh the predictions of an expert
player and because action is fulfilling
the predictions it will look as if it's
an it is an expert
player and here are the um the Learned
trajectories um at the highest level of
different episodes that we'll see uh um
shown in in an alternative format uh in
the next slide so here's the initial
starting episodes little events here uh
the red dots correspond to reward so you
know first hit second hit and then it
falls into a nice little orbit and this
comp completes um this orbit
indefinitely by in it's the way that it
has learned so remember it's only been
trained on random data so this
particular style of play is unique to
this agent but it is sufficient now to
play with 100% um um uh performance um

simply because it's now going round and
round its orbit very much like the wing
flapping her dug flapping her wings um
these are just different ways of
portraying the same um the same Dynamics
and the same the things that have been
learned um the first eight frames and
then subsequent frames um of expert play
um this is the usual format showing the
posteriors at different levels the
posteriors over the states and the
posteriors or predicted posteriors over
the paths and then rewarded players are
function of the uh free energy at
different levels here illustrated by the
by the red dots every time there's a hit
up until 512 time frames here and here's
this rather long orbit through 50
episodes um each four uh time steps in
in length that subtends this kind of
play this is an interesting one and
speaks to the fact that we've actually
switched on inductive inference at the
highest level um so what does that mean
well
what we've done is just um um
basically take each of these highest
level
States worked out the sequence that it
encodes as by um um Computing or
projecting down through the model and if
that sequence includes a hit um we said
that's an intended State that's the kind
of state that I want to work towards
through the successive um four um time
step
epochs um
and if I want to be at some point in the

future in one of these intended States
in this instance rewarded States or
episodes um what states must I avoid in
order to preserve or conserve the
possibility of getting in or getting to
one of these intended States um
basically that's the technology behind
inducting inference and it's sort of
summarized here in terms of the latent
States of uh you know 50 or so of these
latent States in terms of how many time
steps would it would it what's the
shortest number of time steps until the
intended State um and basically um we um
as we move from state to state to state
we want to keep in the white areas here
avoiding these dead ends if you like
from which there is no possib
of reaching an intended State and it's
really trivial to evaluate these um
areas that you have to avoid because you
know the probability transition matrices
so you just multiply it by itself a
given number of times and you can work
out can I get from this state to this
state and if you can't um then you get
this um black area here and you know you
have to move from here to here to here
to here to get to um your um your
intended state so that's inductive
inference in brief and we've applied at
the top level so we just the you just
told the agent this world um is under
your control imagine you know imagine
what you want um noting that these
particular episodes or events contain
you are things that you should pass
through or should experience um and then

deploying inductive inference to
basically predict what will happen all
the way down at the bottom and then
using reflexes to make that happen
um we can apply exactly the same um
technology or the same procedure to
different kinds of games that a slightly
more um complicated game um me to
emulate breakout um where we've
introduced uh certain stochasticity in
terms of the initial conditions and uh
sticky action but the the overall
behavior is the same it's just basically
choosing paths through episodes into the
future given what knows that can be
achieved in order to get to uh rewarded
States so um
effectively what we've done is create um
an attracting set of episodes at the
highest level of these scale-free um
um generative models where all the
allowable paths and paths now in this
generalized State space or event space
lead to reward um we can go a little bit
further and just um add in uh various
episodes by merging uh or assimilating
new data new training data if and only
if um the path actually leads to an
existing um part of this attracting set
and I've just Illustrated that here um
in this final example so this is the
path or the orbit or the attracting set
that was learned at the highest level
by the um the ping pong playing uh agent
but now we can um fill in um fill in
this um effectively um allowable
Dynamics to include all states that lead
to a path on the orbit um and or thereby

all paths will eventually lead to reward
or to roam so we can augment the model
of expert play with paths that lead to
an orbit namely this attracting set um
and paths that lead to event that lead
to the orbit um recursively and get
quite a dense set of allowable
transitions that we can be guaranteed
always keep within the attracting set
that will contain these preferred States
or um states that we have um constraints
uh where um we have constrained those
states that are not visited simply
because they cannot be recognized and
that's it thank you very much
indeed amazing Carl thank you
awesome okay Aron if you'd like to lead
in with a
first question and Lance too thank you
but go for R
first uh yeah thanks very much Carl um
so my first question is um based on a
sentence in the paper which talks about
any model being sparse must be
renormalizable could you say a little
bit more about why that must be
oh I can't remember give me a
clue uh I I'm presuming it has to do
with Markoff blankets and yes yes yeah
that's as far as I got I see yes no I I
remember now um I it may have been in
reference to some work done by Dalton um
showing that with probability one
provided the system is large enough
there will be markof blankets and as
soon as there are Markoff blankets there
is now an opportunity to take Markoff
blankets or Markoff blankets in the

spirit of um that applying the
renormalization group that that I
started off with um so as soon as you've
got um as as soon as any um sparely
coupled random dynamical
system um can now support a um a
particular partition into Markoff
blankets you've now got the opportunity
to take Markoff blankets and Markoff
blankets and then apply that
renormalizing procedure that I said
before so all you need to posit is any
system um that has a Markoff blanket at
any given level and as soon as you've
got that you've got Markoff blankets all
the way up um not necessarily all the
way down but um you certainly got them
all the way up um the the physics
approach would would actually um also
actually um require for for there to be
markup blankets all the way down as well
but that's that's that's another issue
um so um what that basically means is
that um all you require is a sufficient
degree of
sparcity that would um lead to the
condition
independences that are definitive of
Markoff blankets and in one sense much
of the sort of um the
um the maths behind the free energy
Principle as it inherits from random
dynamical systems or uh in physics say
Lan um equations is simply
showing that a sparse coupling a sparse
causal coupling in a physics or control
theoretic sense not a philosophical or
Granger um or um cause effect structure

sense but in the physics sense of a
coupling so the the change um this state
um influences the rate of change of that
state so that kind of um which could be
written as an influence graph so if
there's a sparse coupling on the
influence graph the equivalent basian
graph or probabalistic graphical model
induced by that sparse coupling um now
uh can be interpreted in the in the
sense of um um Pearl's Markoff blanking
um so all you need to get to the
application of the renormalization group
to markof blankets is sparse causal
coupling hence sparse Rand random
dynamical system that is sparely coupled
just out of interest um the spin block
um the motivation for the spin block
applying the spin block um or blocking
Transformations that basically comes
from the assumption that um many systems
that people deal with both in computer
vision and in uh and in um idealized um
models in physics um has an exquisitely
sparse distribution and that's because
the causal inferences are all local so
as soon as you in um you look at a a
coupled map l ltis or a globally coupled
map you'll notice let's take the ltis
for example let's take a sort of
idealized um uh um um Crystal uh so what
you're saying when you write down a ltis
is I'm only connected to my neighbors to
my more neighborhood or to my Markoff
blanket um that means that in a large
latis uh where I could be connected to N
squared neighbors I'm only connected to
1 2 3 4 5 6 7 8 so a fantastic AI small

almost empty set of possible connections
and I use that phrase because there I
can't remember who said it now but I
really liked it the notion that if one
looks at the connecto in the brain in
terms of the connections it's almost
empty you know the number of actual
connections you know white matter
connections external connections and
indeed intrinsic um um exteral
connections are almost zero in
comparison to the total number of
possible connections between the 10 to
the 11 or whatever uh neurons in the
brain so that's not surprising and I you
it's one of those nice examples that the
the anatomy the functional architecture
of the brain recapitulates the anatomy
of the lived World in it has this
incredible sparcity um um and that just
means um the universes in which there is
um the action at a distance is quite
rare um um means that that you you're
just interacting you're just coupled to
your neighbors and that's and that gives
you if you then look at the mark of
blanket structure or the particular
partition for these lattice structures
you get to the Sim block um blocking
Transformations as a way of carving up
nature you know in terms of little
patches or Markoff
blankets I do have a followup question
on that but then I'll hand it back to
Daniel or Lance um so the ltis example
is really interesting because I think
there's a symmetry there that maybe we
lose when we talk about blankets that

only have one parent so in a normal Ice
as you say you're just connected to your
eight Neighbors in 3D
space when we looked at the spin block
Transformations before we were going
let's look at this square and everything
in this square has one parent but the
square along is connected to a separate
parent so now if you're at the
boundary we are breaking some symmetry
there yes technically or practically a
very good point um it doesn't really you
it doesn't really um
uh have any material impact upon the the
the algorithms or the partitioning um so
just practically what happens is you you
you you you just take your tiles and
then you have a half tile or a partial
tile um until you get to the boundary so
these these groups in the grouping
operation or the or the blocking
operation are not necessarily the same
um you know they're not the same size um
and indeed in uh um current um work
we're actually dealing with sort of
multiple streams so you can have sort of
audio files and video files separately
tiled and then converging right right at
the top um ideally what you'd want to do
is is
um not use this um um blocking
transformation but to try and identify
the Markoff blanket structure at the
particular scale in question actually
use the Markoff blankets based upon um
well there are a number of ways of doing
that you can either use a markof blanket
Discovery algorithm of the kind that

Jeff Beck um Works upon um or you can use um empirical um analysis of the um the dynamical coupling um uh by estimating effective the
jacobians you know for those people are interested there's a worked example of that in the particles and Parcels in the brain paper in network Neuroscience um it's really founded in terms of data analysis but you can use the same principles to um in to identify a particular partition from a large number of um um States or boxal or or or or or pixels um um you in three and 2D um respectively um there's another thing though that I think your question speaks to
um which is um you know you've got you got you know one block of four and then another block of four and then um because of the sparcity that they're separately U modeled at at the level above so I thought you were going to ask how in Earth now do you start to repair the obviously rather facile assumption that you've got just local action
because of course in our world there's lots of action to distance you we are um talking to each other over over over thousands of miles or Kil kilometers and you know we have Vision uh so all the and we have uh linguistic communication via sound so you know it is interesting that we have actually violated
much of the local action if I was a virus this would be perfectly okay because the only thing that's really going to affect me as a virus is the

things that actually are in molecular
contact with me but for you and me um
and the worlds in which we live in um we
can't there is no such thing as just the
action distance now starts to really
matter um and effectively because the
two um latent States generalized latent
States generating stuff um initial
conditions and paths for the two blocks
below because they themselves now
constitute a single uh contribute to a
single um latent State at the at the
level above as you move deeper and
deeper you start to now
repair the simplifying Assumption of
conditional Independence between the
groups so it is at the higher levels
that you now become you're able to see
these action this action at a distance
these coupling your what what happens
over here really matters in terms of
what's going on over here but you'll
only ever see that you only have that as
part of the generative model right at
the top which is why all the inductive
inference is implemented right at the
top level where where you can see
everything um so that it becomes you
know if you like you are
now you are now in a position to um um
model and thereby generate appropriate
predictions
that rest upon um these long range
dependencies um and of course you know
us using the word long range I mean in
the sense of the the coupling distance
uh you you in a dynamical sense is that
what you had in mind uh that was exactly

what I had in mind thank
you thank youon
Lance do you have any remarks or want to
add a question or I can yes please go
for it yeah I have lots of questions uh
thanks a lot C for the talk so I'm just
wondering about this all roads lead to
Rome slide that you had at the end and
I'm wondering whether you have some sort
of curse of
dimensionality as you increase the
dimensionality of the system so it
sounds to me like the manifold of expert
play because um because you're taking
temporal trajectories that's going to be
one dimensional
manifold now as the state space of the
game um increases in
dimensionality you're going to have an
ambient space of trajectory that's much
much higher
dimensional and so it sounds like the
amount of
trajectories they need to you know
construe to bring me back to your um
onedimensional expert manifold is just
going to grow and grow and grow and it
sounds like it might be hard to fill
that space
so could you could you comment a little
bit on that and whether it's a
problem right yeah in general in my
world you know these things aren't
problems but they're challenges um and I
think that challenge is yet to be met to
be met um so what I imagine people will
do um is that you will um ingest
um extra

episodes or
sequences that bring you onto your
attracting set up until some bound um
and then you will switch off active
selection by which I mean the structure
learning um so you stop growing the
model and then you switch on parameter
learning so that now you're forcing the
Transitions and the um expression of
various General Iz states to accommodate
variations around the training data that
you have been previously exposed to so
that fixes the size of the tensors and
hence the dimensionality of the state
Bas State spaces in question um um but
will in principle now accommodate by
preserving by using sort of um Active
Learning in the technical sense that you
only update a parameter if it um does
not uh increase expected free energy or
decrease Mutual information
um then um you you you're now in a
position to become more robust um and
perhaps I should qualified this answer
the whole point of this adding in extra
if you like insets onto an attracting
set is just to equip the agent with a
certain robustness should itself find
itself outside or off that attracting
set so um I know Lance knows this but
I'll just make this explicit for for
everybody else um this kind of learning
um and state action policy um
implementation is very much like
learning to ride a
bike um so you you basically um just
accumulate and remember those
experiences that were successful so for

example I you know managed to um rotate
the pedals by one cycle and I don't fall
off and I remember that and then I
another go and I wait until I've
actually done two cycles and I don't
fall off and I remember that I forget
everything every time I I I fall off and
then I keep on going until at some point
I connect back my last cycle was
identical to the first cycle and then
I've learned to ride the bike but
because I've forgotten all the falling
off I can't recognize or know where I am
when I when I've fallen off so I've got
a very little expertise that will not
allow me to recover if I fall off the
edge so another example which makes it
slightly clear I think is like mountain
climbing you know if I want to be an
expert mountain climber I cannot learn
by my
mistakes provided I'm climbing a
sufficiently high mountain should I fall
off it then I will die another example
here is learning how to cross the road
as a child you know you cannot learn by
um um experiencing being run over but
the cost you um you pay or the price
that you pay is you can't recognize when
you are being run over um and that I
repeat lends a certain brittleness and a
lack of robustness to agents who may be
exposed to Danger that doesn't actually
terminate them so putting this
robustness back in by growing the um by
growing the attracting set or basically
putting these insets on um in this sort
of generalized um phase space um you

know it's just to make it more robust um
in numerical studies it it does seem to
um converge it doesn't seem to grow inde
definitely you so those are likely
violations are moving off the attracting
set are themselves quite small
relatively small in number um yeah
practically speaking uh if if you've
control the sources of stochasticity or
unpredictability so it may not be always
the case that you need to have an
explicit bound on the upper the size of
the um the very high level
tensors if if you do then one has to now
think of a principled way of making that
upper bound an emergent property of um
um expected for energy minimization it
may be that you know you violate now the
mutual information that certain rare
events are just so rare you're actually
compromising the mutual information by
including them as a particular option um
or a particular column of of a likely
likelihood mapping or yeah but I don't
know but these are all really
interesting questions to
pursue thank you Carl all right Arun go
for it
um yeah so following on from that I
think there an interesting question on
the uniqueness part of ingesting data
and the question is are there many more
ways to fail than there are to succeed
so if you're passing your data into a
sort of Maxwell demon Maxwell's demon uh
with a gate saying accept this data
reject this data if this uh improves my
Mutual information if there are many

more ways to fail but then get back onto
the
a successful path but there are only a
few paths that actually keep you there
would it not make sense to just
continually accumulate these sort of
recovery
States because the mutual information
will keep increasing and then you stop
there then you parameter you learn
parameter wise because we're not
including the sort of Prior distribution
of How likely you are to be succeeding
at any given moment because with the
uniqueness thing you don't know oh 10
times out of 10 I'm cycling or like nine
times out of 10 I'm cycling and one
times out of 10 I'm falling off the
bike yeah I think it depends
um again I don't have any explicit
numerical analyses or analytic analys
maatic analyses of this but
intuitively um I think during learning
there may well be um um good mileage in
in gesting failures um or roots to
success if you like or to the attracting
set that that actually Traverse regimes
that would normally be excluded by my
constraints or my PRI preferences um you
know just taking the view of the role of
of the sea tensors or prior preferences
from the point of view of physics what
that basically is saying is it's
actually carving out vast regimes of
State space latent State space that you
should not be in because the it would be
uncharacteristic of you to be in these
things so it it's really not so much

about the rewards and the preferences uh
it's really what you where you shouldn't
be and that's what defines the
attracting set um however before you've
learned that attracting set it may be
useful to include constrained regions of
State space that offer a root into or
onto that attracting set so I can
certainly see that that would be a
useful device but notice that once
you're on the attracting set you will
never visit the constrained
regions because you now keep yourself on
the attracting set um um and therefore
if you've encoded those but you never
visit them your Mutual information will
actually
fall so what that means operationally is
what you'll probably do is have this
sort of slightly over inclusive
structure
learning learn how to cope with all the
necessary sources of Randomness so that
your attracting set is sufficiently
robust and you do actually visit from
time to time but not um not never um s
of excursions from the attracting set
are drawn back onto the attracting set
um and then use basing model reduction
to eliminate all the states that you
don't want to recognize simply because
you never go there um you know once in a
you know when I was younger I would
certainly be able to rec what it's like
to fall off my bike but now as an expert
bike rider I really don't want or need
that so you know and that's just
basically getting old and wise and you

know implementing bising model reduction
to remove the Redundant parameters which
would in this instance just be the
columns um and um
Associated slices or or rows of the of
the transition matricies so I think you
could perhaps another way of then ADV
answering Lan's question is yeah you
just keep um
accumulating um but um start to engage
Bas in model reduction at some point you
should reach an equilibrium which will
be the attracting set that allows you to
survive in the character states that
Define you in this particular
environment with this particular
volatility thank you
awesome okay I'm going to bring in a
question from the live chat
okay Javier wrote so stochastic plus
dynamical probabilities learned by the
agent itself are scale-free will this
hierarchical learning also be applied to
the goals of the agent different
abstractions granularities of goals
[Music]
self-learned my answer would be yes but
I'm going to I'm going to ask Lance to
to to answer that one
oh that that's
cheeky uh yeah I would say for sure um
because typically when we have when we
oper operational operationalize sorry
the goals in our gen model they need to
have the same structure at our Gena
model itself so I would imagine then we
would have a hierarchical renormalizable
goals which kind of makes sense

cool okay I'll bring in oh yeah go for
it Carl I was just going to say um you
know just to um unpack what Lance just
said in terms of practical things you
know if you're trying to um you're
trying to say um what my goal is is to
go uh is to have dinner um that
naturally unpacks hierarchically into
I've got to move from the living room to
the kitchen then I have to find the you
open the fridge then sorry I have to
find the food which then unpacks into I
have now open them and all the way down
to to the finest finest muscle movement
so almost by definition um your know
intended
narratives can always be um decomposed
in this scale-free way you um there
there's a this is very similar to sort
of um motor chunking and hierarchical
decomposition in in um um theoretical uh
motor control for example and you
probably find it a lot in in robotics
there's a global long-term plan that
that provides empirical priz of
constraints on shorter little chunks
that themselves provide constraints on
shorter and shorter chunks and that's
exactly the um if you like the the
architecture implicit in these scale
free
models awesome and totally ties back to
the synergetics
okay I'll ask two questions from ml Dawn
he wrote how can we inject the same
separation of time scales by imposing
smoothness constraints on say the
parameters as opposed to playing with

proportional update rates at different
levels and this relates to something
Arun and I talked a lot about in the
zero and leading up to it is
identifying which
Windows to
use you could imagine picking the right
separation of time scales if the classes
on campus really do shift on the hour
then the hour course graining is going
to be great and effective but the 59 or
the 61 minute might Alias you with the
real variability of the system and so
there's not necessarily even a smooth
landscape for determining the windowing
or the binning of coar graining and coar
graining may have dependences on each
other so how do you go
about playing and finding viable
approaches and robust ones amidst all
these different levels of of meta
optimization that are opening
up I'll take this one can take take the
next one um then that's a really good
question because there is an implicit
separation of time scales between
um so vanilla implementations of
inference and learning however in these
schemes there isn't um so the the know
the um the Matlab version or
implementation um of um active inflence
under these and the fact all uh markof
decision process models um unlike um the
um procedures that you might find in
data analysis where you acquire you
basically do some filtering on the on a
an Epoch of data and then at the end you
update your parameters and then you

rinse wash and repeat um in um in active inference um simulations and also in some um applications of what's called generalized filtering you have continual learning so the update of the parameters actually occurs a at every time step um at each level of the model so the likelihood parameters at the lowest level are updated every time that routine is called um so there is continual learning the learning um is yes you're absolutely right is smooth in the sense that you're slowly accumulating drish parameters but the tempal scale of that learning is the same as the inference um and you could also argue well could one also put model reduction um into play at the same time scale so basically can you at every time point at each level in the model um identify redundant parameters and remove them uh using Bas and model reduction in effect that's what's done uh using Active Learning what I said before Active Learning will only proceed um if the expected free energy of mutual information improves that's actually also implemented so the the kind the the the rate the rate at which it looks as though things change um between inference learning and model selection is certainly distinct I don't think that's quite the same kind of separation of time scales that is implied in the renormalization group that's as you were intimating uh Daniel exactly specified by the number of ticks or updates at level n per single update

date at the uh level n+1 um and that is
I think you the you the key question
you're asking how do you get that
right um and of course the answer to
that is the model that has the greatest
model evidence or smallest free energy
or greatest marginal likelihood for the
kind of data generated by this
environment in this world so um are
there any um sort of useful prizes you
can can put over this um well what one
one useful prior is um that um things
are changing as uh to to well one PRI
which is actually used in operation is
you take the lower limit on on What's
called the the temporal RG flow which is
basically the the length of the window
at each and every level um at the moment
that's two what does that mean how can
you motivate that well it's the most
expressive in the following sense
but if I um just imagine um what imagine
what this dis discrete discretization of
time means from the point of view of
classical Notions in continuous State
space like position and momentum or
position and
velocity if um if I use a
a um a scaling of two which is basically
means there are two time steps or every
time step the um the level above what I
effectively doing is um um modeling two
time steps for every state at the level
above and that basically means I can
model the initial condition and the
subsequent condition basically terms of
the change which of course is the
velocity um and what does that mean for

the level above that well it now means
that I'm modeling the change in the
velocity so I'm now modeling the
acceleration and then so on up until um
you know the high order of motion so
what that means is by taking this
limiting case of T the temp temporal
horizonal the the you the ratio of time
time steps from one level to the other
level um as two means that I got a very
expressive model that can handle um can
recognize um things that change very
very quickly and their acceleration
their jerk and all and the implicit um
context sensitivity that that that gives
you if I didn't do that I'd have to
assume that I C grain over long periods
of time and
therefore the path is consistent during
that time now that sometimes is a useful
approximation and indeed there's a whole
industry called you know called linear
switching linear dynamical systems that
makes that approximation it means you
can um approximate all the nonlinear
complicated high order motions
in some evolving pattern or system with
a series of um um linear
trajectories that basically go you know
that intercede between a series of
points at which you which you which you
switch the problem with
that is that the length of the linear
trajectories for as for example I'm
committed to one Path U so one slice of
my B Matrix for the next next eight time
steps the problem with this is that you
don't know when you when you deviate

from that transition Dynamics so then
you get into the game or perhaps I could
make my scheme reactive so sometimes I
do three steps sometimes I do eight
steps sometimes I do one step and I have
some criteria for saying your
predictions about my trajectory have now
been met please um I'm going to tell you
give you the evidence that I've been
pursuing this path you up dat your priv
level above and tell me what you expect
me to do next so this now speaks to um a
particular kind of reactive message
passing um that starts to have
implications for the computer science
and the sort of CR the crud um um or um
as um as I think we've rehearsed before
um The Institute the the original
conception uh behind um
the actor model in computer science from
the 1990s so it's all about reactive
message passing I will send you a
message when you send me I will return a
message to you if you send me a message
um and what this um what these um
renormalizing time models have and I
should say that this also applies in the
absence of the the spatial
renormalization so any deep temporal
model will have this aspect what that
means is that from the point of view of
Any Given
level I will um be responding to
messages more quickly from my
subordinates than from my
superiors so once in a while I'll get a
request from a
superior my response to that is to ask

my juniors to do a series of things they
report back to me and when they report I
give them the next bit bit um request or
guidance uh until they complete their
job and when all my juniors have
completed their job of course they be
normalizing specially normalizing
context I'll have a whole group of of
Juniors to worry about um then um I have
gathered enough evidence for me to then
return the the message to my to my
Superior so you get the in terms of the
reactive message passing in this sort of
actor model of um of Compu to message
passing um you get the the separation of
temporal scales inherits from the number
of messages or re um requests and uh
responses um um exchanged with the
people lower down in the hierarchy
relative to the number immediately
subordinate
um and in principle they don't have to
be fixed so there could be a
criteria um before I pass the message
back back up to the to the higher level
I think that's Daniel probably what you
were getting at um I'm not there may be
somebody who's done that um um but
I'm I haven't done that and I I I don't
know uh I can't give you an example of
somebody who has done that you may find
you may find somebody um in Bert de's
group um who sort of you know some of
the world experts um in reactive message
passing who have considered these kinds
of things
um you know you in in principle um you
know there should be a way of

um each level testing whether I have
fulfilled the predictions of my
superiors um and now it's time to get
more guidance you know uh based upon the
energy awesome the organizational
technology I did didn't know that my PhD
needed okay I'll ask a question as we
get close to the end and and this will
be for for
anyone ml Dawn wrote how does the
training time of active inference
compared to that of reinforcement
learning and how does their scalability
compare and it would be interesting just
to hear anyone's very
2024 takes
where are we at with that entire Arc
from problem
conceptualization on through the
training and the inference how do we
even grapple and
describe with similarities and
differences in the training time and
resources and all of this
that's definitely Lance for for at least
five minutes I
think I'm I'm a bit incapacitated in
terms of talking um I think this is
actually a good question for Aron um but
yeah I mean I can say a few
things um so uh scaleability in actor
inference I would say it has been a
challenge and there has been um much
less man work hours that have been put
into that than in reinforcement learning
now for the past few years we we've had
a lot of work um a lot of it led by C
actually um trying to scale the

structure learning in active inference
and and I think we've done a lot of
mileage on that so there's been the
supervised structure learning paper
there's this paper uh there was also
some interesting work um presented at
the international work sh and active
inference um I personally think there
are still some uh challenges ahead and
and I think uh one of them is is the one
that we just talked um a few questions
ago about you know if you if you learn
an expert orbit how do you get back on
the orbit as the dimensionality of the
system increases so this is definitely
um one
challenge um now I know much less about
reinforcement learning so um yeah maybe
Arun has more things to say there um but
but what I will say is a lot of
reinforcement learning is powered by
Deep neural networks when we use deep
neural networks we sacrifice uh the
interpretability of our generative model
this is one of the strength of active
inference is having more
interpretable um Dynamic Cal models
where we can actually peer into them and
see okay well this representation
includes this this other one includ that
and so
forth um the fact though that
reinforcement learning typically uses
deep neural networks means that it has
been much more scalable uh just because
there has been so much work done in like
training deep networks this is a bit of
my uh my perspective I don't think it

fully answers the question
though I can come in with a little bit
on that but I must admit that I've not
trained any reinforcement learning
agents myself I worked with people who
have uh I'd agree Lance that the
infrastructure generally is much more
advanced for training RL agents for
doing stuff uh and completely agree with
as you say the explainability and the
interpretability of why did an RL agent
pick a particular action is very opaque
uh and I think the deeper those networks
become the harder it is to understand
what's going on in the hidden layers and
uh generally why they do what they do I
think with Act of inference yes we do
have more
explainability but I do wonder if we go
the far structure learning approach I
think Carl had a particular uh comment
earlier about looking at the highest
level um identifying the desirable
States learned by the
agent from what I understand the
identification has to be done by a human
they have to look at the model that's
been learned by this uh renormalization
approach look at those top levels and
then say ah what do these levels mean so
an example in the paper is the Mist uh
classification it's the case of a human
being looking at those top levels and
going ah these top levels each
correspond to a digit in the handwriting
task
so we are uh sacrificing some
explainability I think by doing this

fast structure learning compared to what
might be the old fashioned way way of
creating uh G models By Hand by thinking
sitting in my armchair and thinking this
is how I might model the world these
structures these likelihoods these
states and when you do that I think just
by hand you have 100% explainability
that model just might not perform very
well so I think there's always going to
be that tension there um but in terms of
how quickly these things
learn um having not trained either I'd
be having a reckon and at the moment I
would say probably an RG um active
inference agent would train faster
simply because it's able to use the
structure in the data much more
effectively that would that would be my
take and I think as well really
interesting thinking about the active
data selection part of this too I know
that there is some stuff in the paper
about the efficiency being both a curse
and a burden but I'll hand over to Lance
and Carl to give more details on that
yeah
Lance yeah just to um echo echo what
Arun said um also so so the structure
learning active inference and in
reinforcement learning is is also very
different um in reinforcement learning
we typically have a a model with deep n
networks that's massively
overparameterized but it has some fixed
representational capacity from the
outside because the structure of that of
that model will be specified and then

typically the model will learn and then
probably you would prune it using some
techniques like Dropout or or other
things if we looked at that from the
perspective of active inference it would
be like specifying a massively
overparameterized model and then pring
it with Bas model
reduction the um but but now in active
inference we have this Bas and model
expansion for example in this paper and
what this Basin model expansion does is
it learns from the get go the minimal
model for explaining the data and so in
that sense it's extremely
efficient and also provably because it
extremes the um the marginal likelihood
or the mutual information
yeah I just wanted to Echo that last
Point that's a really important Point um
so Lance is drawing the distinction and
there are many distinctions you can draw
with um deep RL um and um deep active
inference read in a more in in its um
generic sense um and that that
distinction at the level of structure
learning or model selection I I think is
absolutely crucial so you
know deep RL starts with an overly
expressive model with a known functional
form um that is deliberately
overparameterized and then shrinks it or
reduces it to um maximize the mutual
information the expected free energy
sometimes they um referred to as a cross
entropy in machine learning so the
reward function now becomes uh there's
no explicit reward in unsupervised

learning it's just replaced by the
mutual information or the cross entropy
um which should be distinguished between
supervis enforcement learning such as
you know digit classification for
example um whereas the you know active
inference um under these re under fast
structure learning is exactly as Lance
says it starts from nothing and stops
when it's minimally complex which means
it's maximally efficient and that
efficiency translates in many guises
through to sample efficiency through to
um um statistical efficiency through to
thermodynamic efficiency and this
matters because from the point of view
of climate
change this is the right way to do it as
opposed to building power stations to
power large language models and
Transformers that need big data so um
the efficiency um on all levels whether
it's the the sample efficiency or using
the right data uh or whether it's right
through to how much electricity do you
need to actually train your model and
how many millions of dollars do you need
to trade a foundation model um all of
these are reflections of the failure of
um machine learning to write in the
efficiency into the ultimate objective
function which is un said is just the
marginal likelihood of the evidence for
this model of of this kind of content or
or
World awesome okay final question and a
very apt one as we head into the rest of
the Symposium so just give any short

thoughts Tintin wrote hi all thanks for the incredible event does the sequence does the sequence in which the training data is fed to the A structure learning model impact what the resulting learn structure would be so the learning curriculum AKA Symposium program challenge well the answer is sorry I'm mindful that Lance is recovering from his um jaw surgery um yeah absolutely um and this is um again just in relation to to to to the last question something I think that eludes uh conventional RL certainly that it rests upon um um classification procedures and um because um the active inference as an application the free energy principle is all about the Dynamics of self-organization and Dynamics means time and time means sequence matters so inevitably at some level the order in which you see things is absolutely crucial for getting the right kind of generative model um that is even in the case um actually of static image recognition because you're assuming that there's no ordinal structure or correlation structure to the presentation say of emus digits you're assuming that they are selected at random as opposed to seeing all the tens and then all the all the ones and then all the all the sevens um so even you in these edge cases um um which are you know the focus of um 20th century RL um you know static image classification for example um your order does matter but they they it matters absolutely acutely

because of course this the order in which you um see um content in um the infast structure learning determines how you populate the um the transition tensors and it's a transition tensors that enable the separation of temporal scales that enables the ability to um encode narratives and plans that far transcend the actual rate at which you samp sample the environment you could also argue that you know this issue has emerged um in a in a strange kind of guys in Transformer models so the attention heads operate on the past so the the where you select the from the past in terms of which token is going to predict the next token most efficiently um again is exquisitly dependent upon and sensitive to the order in which you train train train these
models So Daniel have you thought carefully about your ordinal approach today I I let the structure I let the structure of the niche and others' constraints provide the first pass and make requests from there so thank you Carl thank you Lance for joining again thank you run for all the contributions with azero Carl great to see you
so next up we'll take a 30 second break and we'll be back with Michael Lenin so thanks fellows see you
soon e